



De-Havilland DHC-8-400 B1 & B2 TRAINING MANUAL

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CHAPTER ATA-55 STABILIZERS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Stabilizers, General

Introduction

The stabilizers give stable attitude around the lateral (pitch) and vertical (yaw) axis of the aircraft.

General Description

[Refer Figure 55-1](#)

There are two stabilizers, a vertical and horizontal stabilizer. The vertical stabilizer is constructed as part of the aft/tail fuselage.

The horizontal stabilizer is attached to the top of the vertical stabilizer. The joint between the vertical and horizontal stabilizer is covered by a removable, four piece composite construction aerodynamic fairing.

The fore rudder is attached to fittings on the aft edge of the vertical stabilizer. The elevators attach to fitting on the rear spar of the horizontal stabilizer.

Stabilizer components are made from high strength aluminum alloy.

The stabilizer has the sub-systems that follow:

- Horizontal Stabilizer
- Elevator
- Vertical Stabilizer
- Rudder

Detailed Description

The vertical and horizontal stabilizers are made of high-strength aluminum alloy. The structures use conventional spar and rib construction. The skins are stiffened with bonded stringers.

The leading edges are constructed of composite material. Pneumatic deicing boots are attached to the stabilizer leading edges.

The vertical stabilizer is an airfoil that stabilizes movement of the aircraft around the vertical (yaw) axis.

The rudder is a primary flight control surface that is attached to the rear spar of the vertical stabilizer.

The rudder has a fore rudder and a trailing rudder. The fore rudder is attached to the vertical stabilizer rear spar, and the trailing rudder is attached to the trailing edge of the fore rudder.

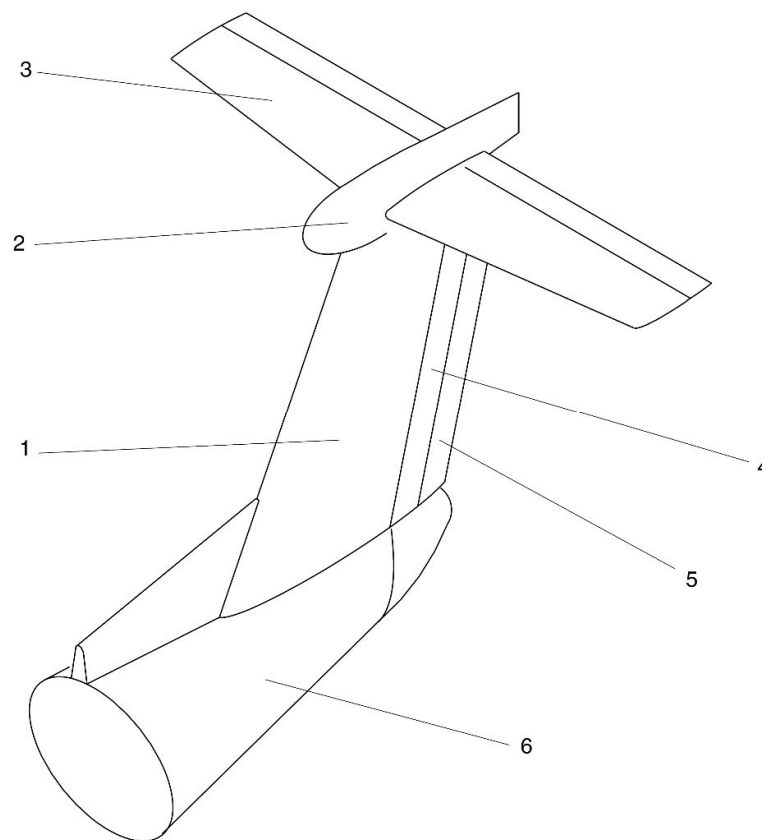
The horizontal stabilizer is an airfoil that stabilizes movement of the aircraft around the lateral (pitch) axis.

The left and the right elevators are moveable surfaces that control the movement of the aircraft around the lateral axis.

Vertical Stabilizer

The vertical stabilizer is part of the aft/tail fuselage. The stabilizer has three vertical spars connected by horizontal ribs, and covered with skin panels.

The vertical spar members are shaped at the lower end to form the integral main frames of the aft/tail fuselage. The leading edge has pneumatic deicer boots installed.



LEGEND

- 1. Vertical Stabilizer.
- 2. Stabilizer Fin Fairing.
- 3. Horizontal Stabilizer.
- 4. Fore Rudder.
- 5. Trailing Rudder.
- 6. Rear Fuselage Section.

Figure 55-1 Stabilizers – General Layout

Rudder

[Refer Figure 55-2](#) & [Refer Figure 55-3](#)

The rudder has a fore rudder and a trailing rudder. The fore rudder is attached to the vertical stabilizer rear spar, and the trailing rudder is attached to the trailing edge of the fore rudder. The fore rudder is moved directly by hydraulic

power. Mechanical linkage transmits fore rudder movement to the trailing rudder. Both rudders are constructed of spars and profile ribs covered by skin panels.

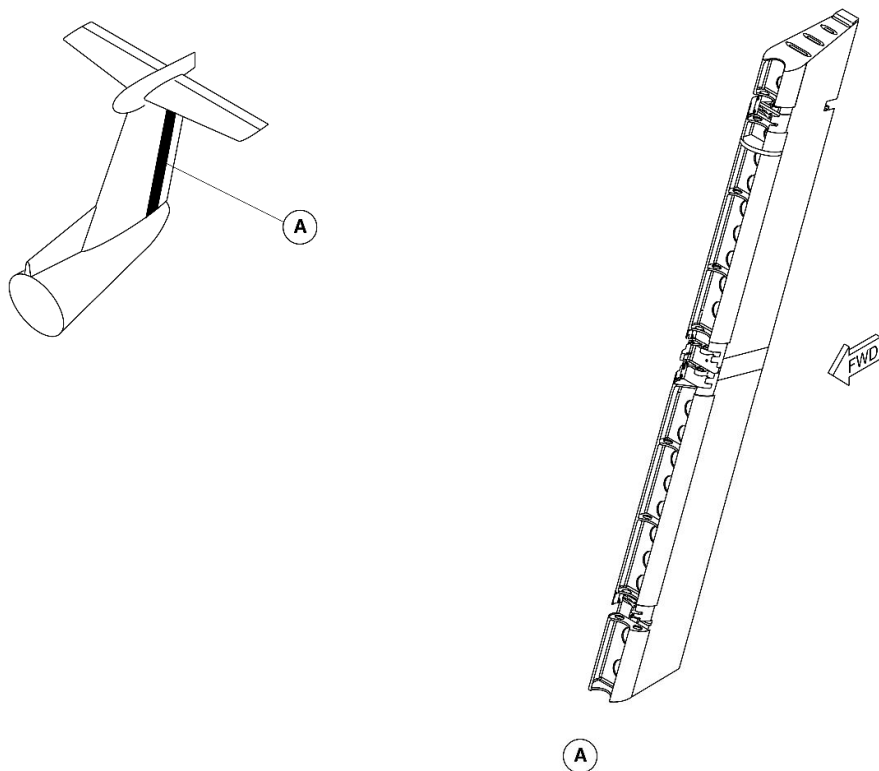


Figure 55-2 Fore Rudder – Location

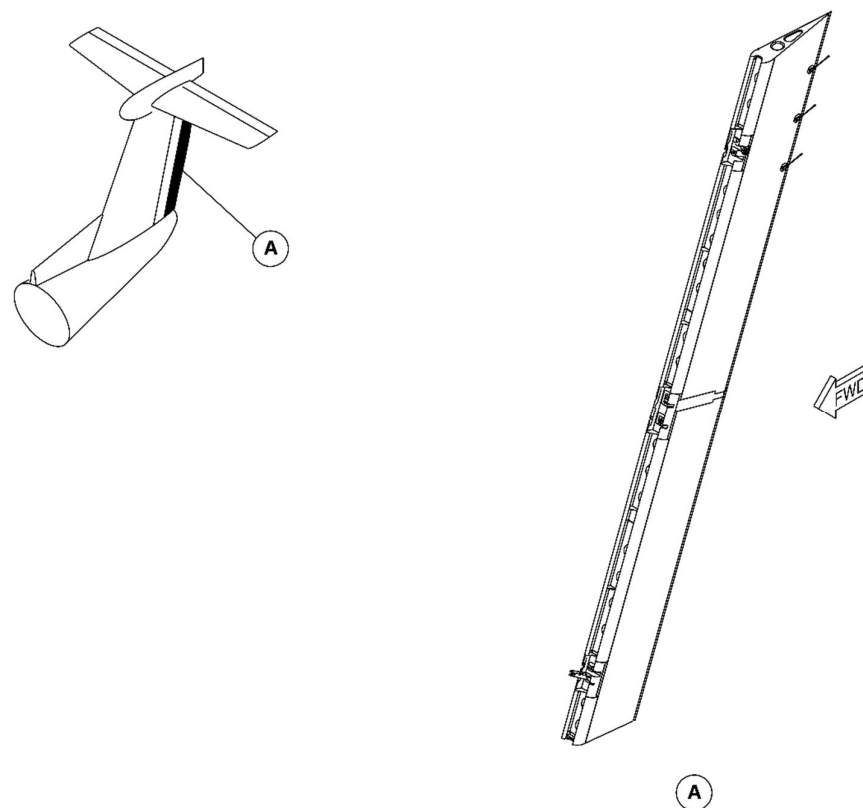


Figure 55-3 Trailing Rudder – Location

Horizontal Stabilizer

[Refer Figure 55-4\](#)

The fixed incidence horizontal stabilizer has a front, mid and rear spar connected by profile ribs and covered with skin panels stiffened by riveted stringers. Each of the three spars has two fittings for attachment of the horizontal stabilizer at the upper end of the vertical stabilizer. Hinge arms for the elevators are attached to the rear spar and adjacent ribs. Leading edges are of a composite construction in two sections per side, installed with locators to prevent incorrect installation, and have pneumatic deicer boots installed.

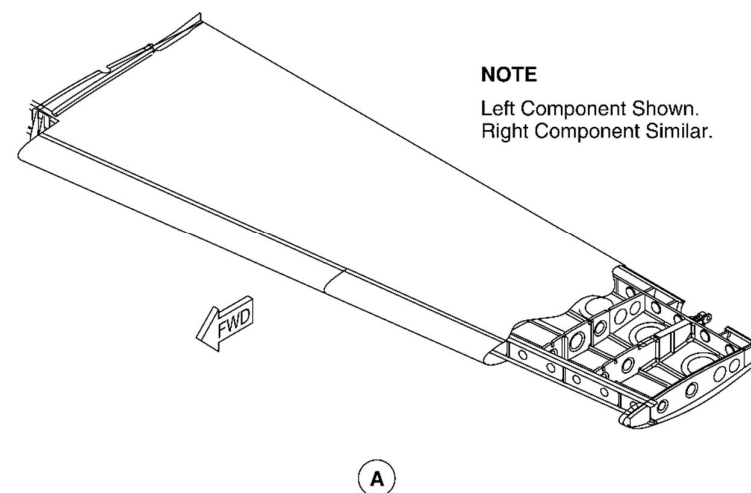
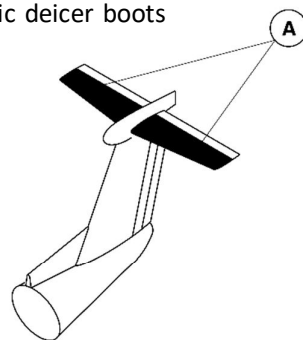


Figure 55-4 Horizontal Stabilizer – Location

Elevators

[Refer Figure 55-5](#)

The left and right elevators has a main and rear spar interconnected by profile ribs and covered by skin panels. The inboard half of the elevator also incorporates a front spar installed with a leading edge sealing strip. The elevators are moved directly by hydraulic power.

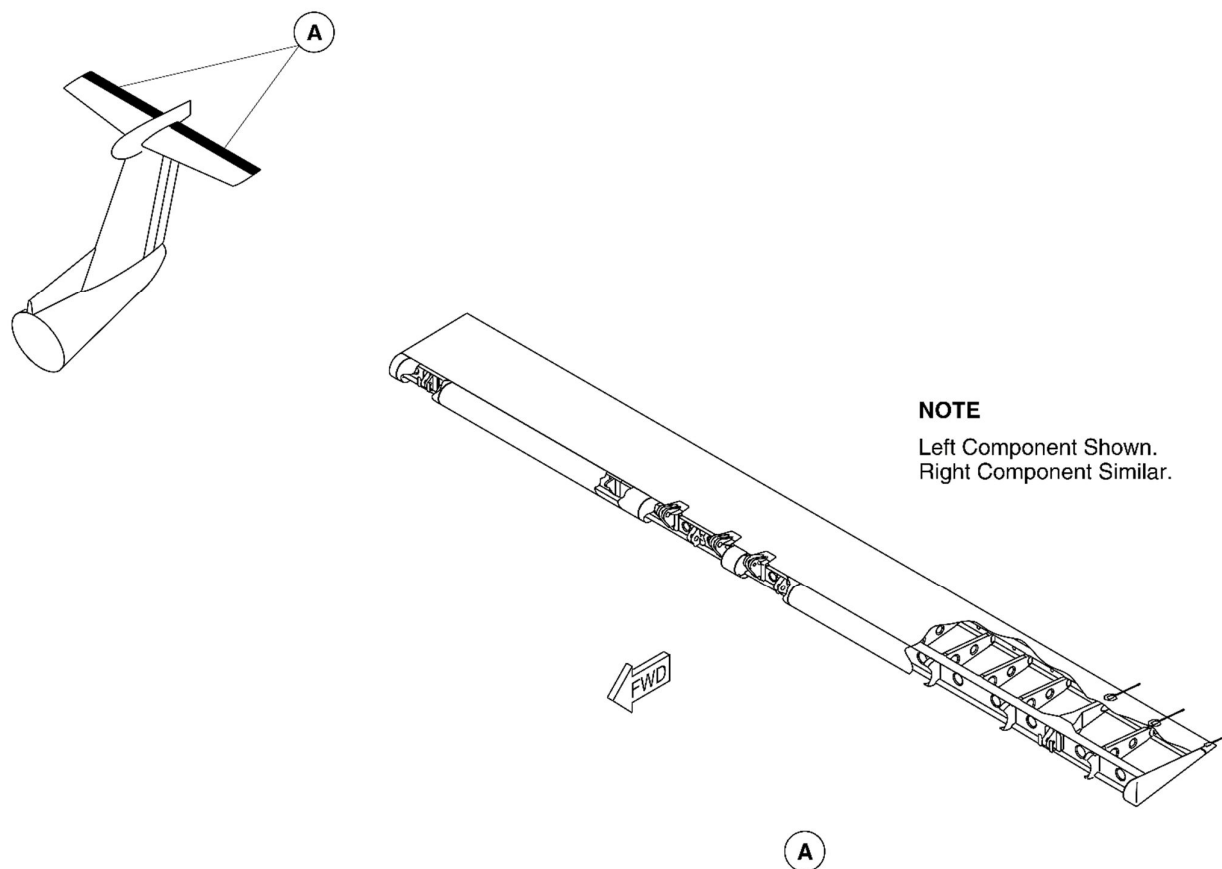


Figure 55-5 Elevator – Location

Horizontal Stabilizer

Introduction

The horizontal stabilizer is an airfoil that stabilizes movement of the aircraft around the lateral (pitch) axis.

General Description

[Refer Figure 55-6](#)

The horizontal stabilizer has the structures that follow:

- Box Structure
- Horizontal Stabilizer Leading Edge
- Tip Fairing Structure
- Elevator Attachment
- Horizontal Stabilizer Access Panels

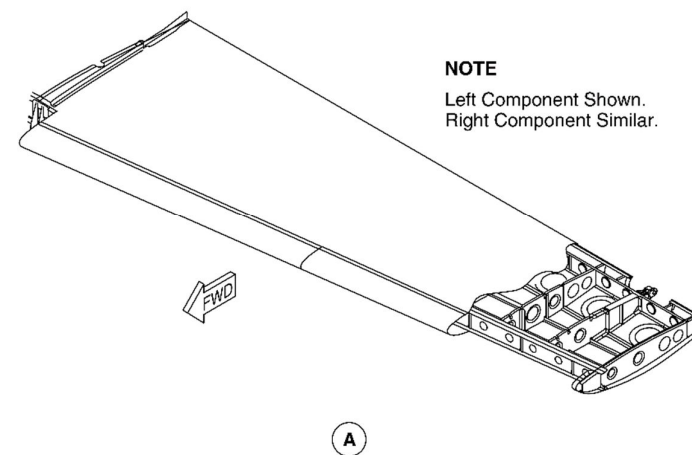
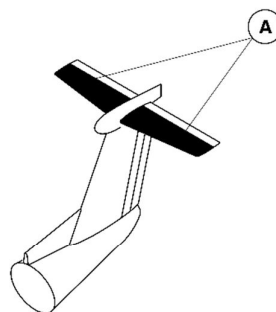


Figure 55-6 Horizontal Stabilizer – Location

Detailed Description

[Refer Figure 55-7](#)

The fixed incidence horizontal stabilizer has front, mid, rear and auxiliary spars. They are connected to each other by profile ribs and covered with skin panels stiffened by riveted stringers. Each front, mid and rear spar has two fittings for attachment of the horizontal stabilizer at the upper end of the vertical stabilizer. Hinge arms for the elevators are attached to the rear spar and adjacent ribs. Leading edges are of composite construction and have two sections per side. The leading edges are fitted with locators to prevent incorrect installation and have pneumatic deicer boots installed.

Box Structure

The box structure is the primary structure of the horizontal stabilizer. It is formed by front and rear spars, and 16 ribs. The top and bottom skin panels are the covers for the box structure. Spanwise stringers give stiffness to the skin panels.

The box structure is made from machined and formed aluminum–alloy materials. The skin panels are made from aluminum alloy sheet, clad on the outer surface, and chem–milled on the bare inner surface. The front and rear spar caps are made from aluminum alloy extrusions. The spar webs and formed members such as some of the ribs and small parts are made from clad aluminum–alloy sheet. Machined fittings and some of the ribs are made from aluminum alloy forgings.

Horizontal Stabilizer Leading Edge

The horizontal stabilizer leading edges are of a composite construction in two sections per side, installed with locators to prevent incorrect installation, and have pneumatic deicer boots installed.

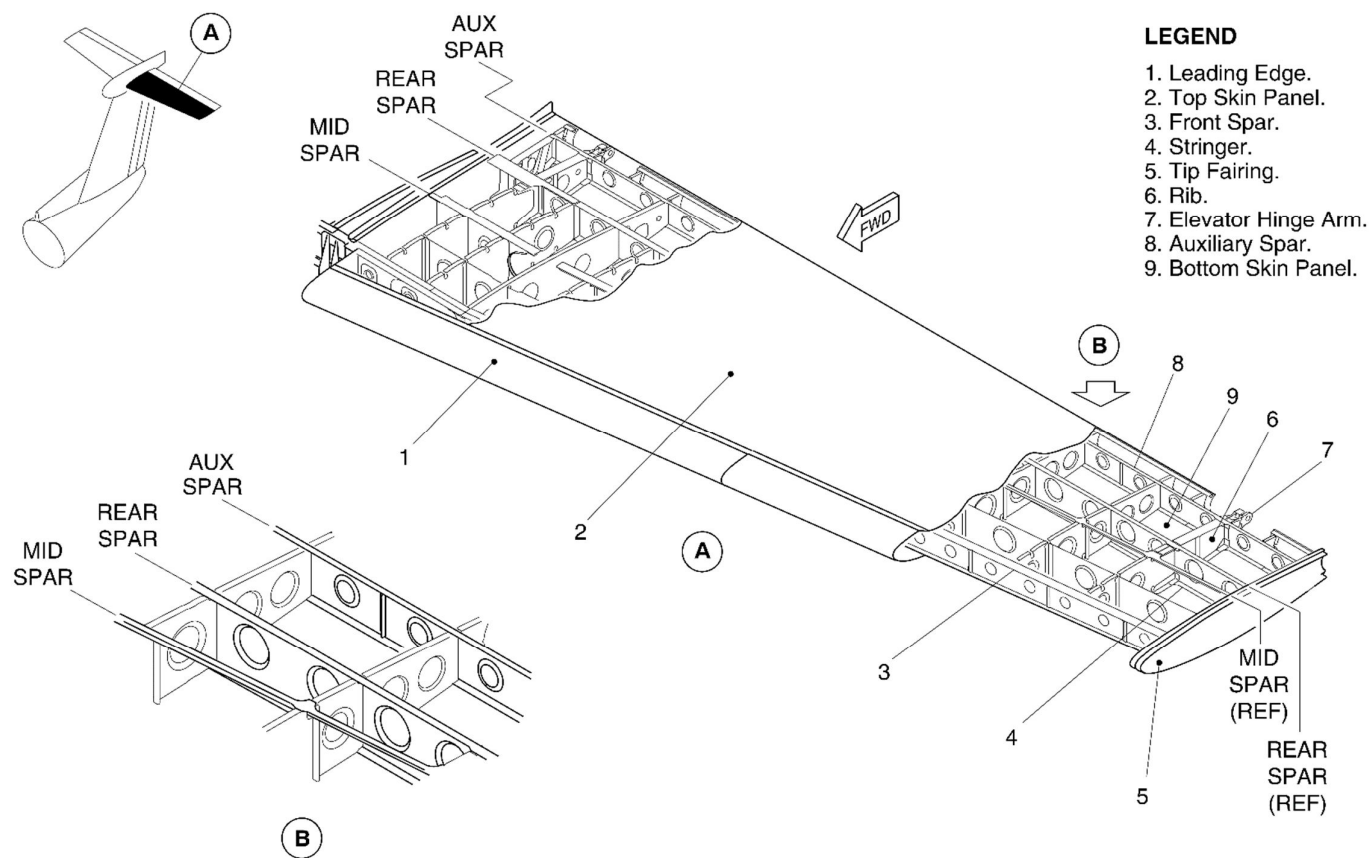


Figure 55-7 Horizontal Stabilizer – Detailed 1 of 2

Tip Fairing Structure

[Refer Figure 55-8](#)

The tip fairing structure is a cover for each end of the horizontal stabilizer. The tip fairing structure is attached to the box structure and to the leading edge assemblies with countersunk-head screws.

The tip fairing is also called as the wing tip. The discharger with discharger base is installed on the strap and the strap is bonded to the wing tip with the adhesive. The dischargers are installed one on each wing tip. The dischargers dissipate electrostatic charges from the aircraft structure to the surrounding air, to minimize radio interference caused by these charges. Each static discharger consists of a resistive coated fibre glass rod with a pellet of composite carbon-based material at the tip.

Elevator Attachment

Four attachment hinges for each elevator are installed on the auxiliary spar on each side of the horizontal stabilizer. The hinges are machined from aluminum alloy forgings.

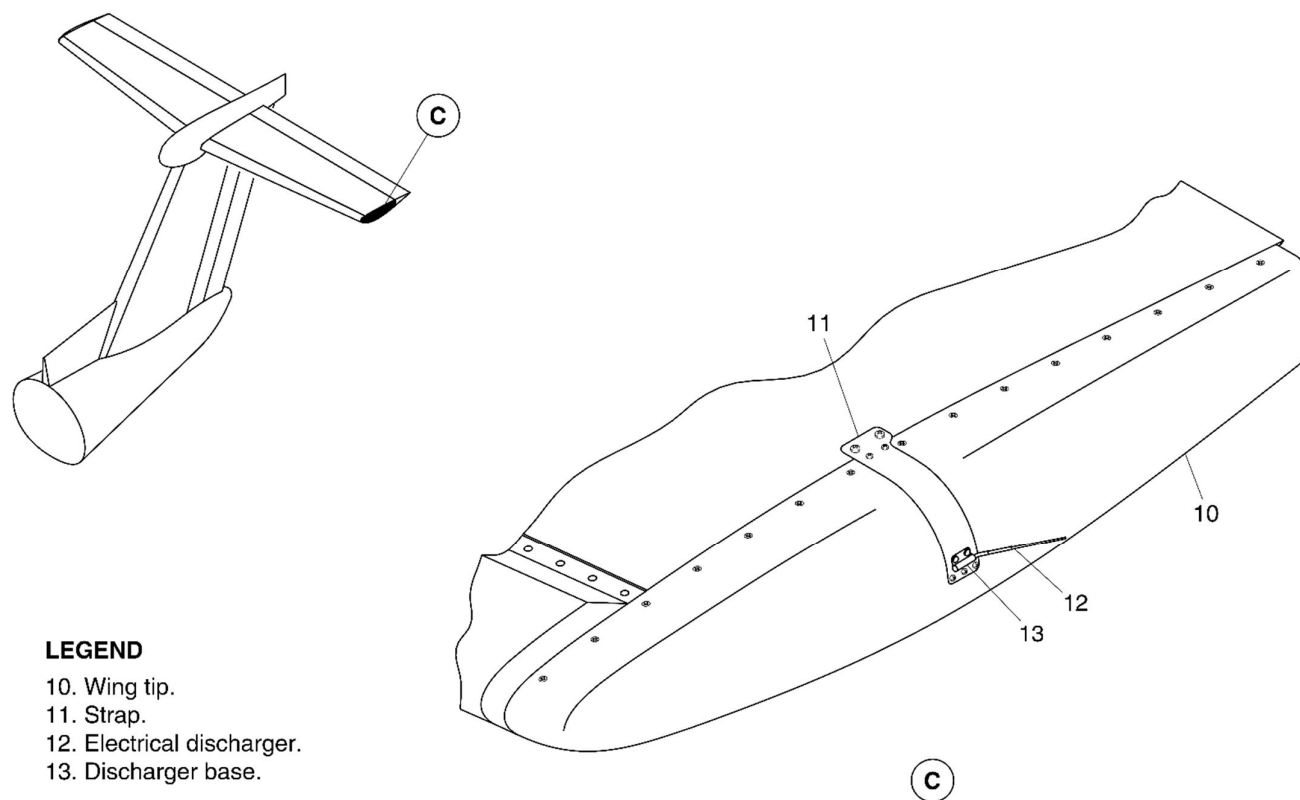


Figure 55-8 Horizontal Stabilizer – Detailed 2 of 2

Horizontal Stabilizer Access Panels

[Refer Figure 55-9](#)

Thirteen access panels are installed on the bottom skin on each side of the horizontal stabilizer. The access panels give access to the deice lines and the elevator control system.

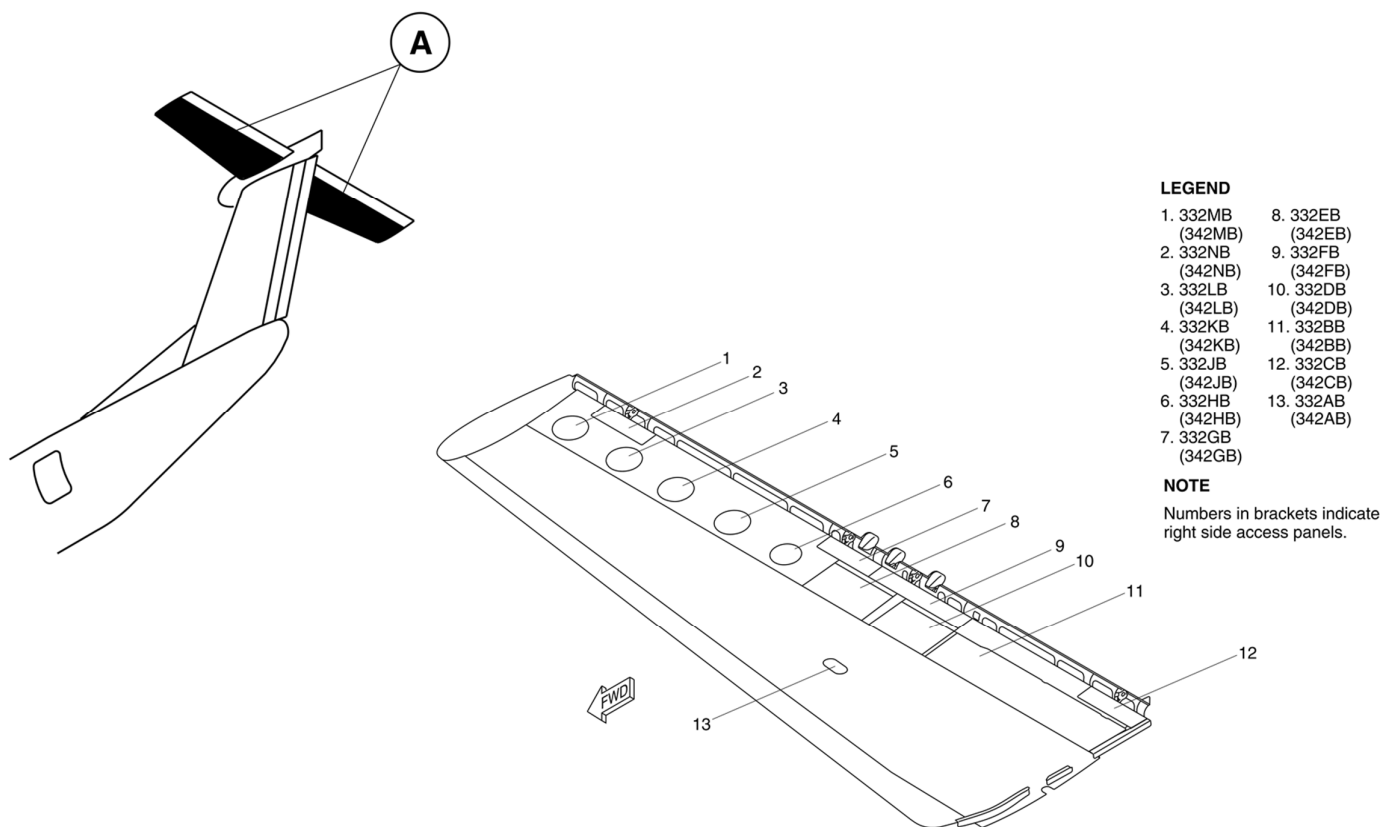


Figure 55-9 Horizontal Stabilizer Access Panel

Elevator

Introduction

The left and the right elevators are moveable flight control surfaces that control the movement of the aircraft around the lateral axis.

General Description

[Refer Figure 55-10](#)

The left and the right elevators are installed on the trailing edge of each side of the horizontal stabilizer. The elevators are attached to the horizontal stabilizer by hinges.

The elevators include the structures that follow:

- Box Structure
- Elevator Attachment
- Elevator Power Control Unit (PCU) and Linear Variable
- Differential Transformer (LVDT) Attachment
- Elevator Access Panels

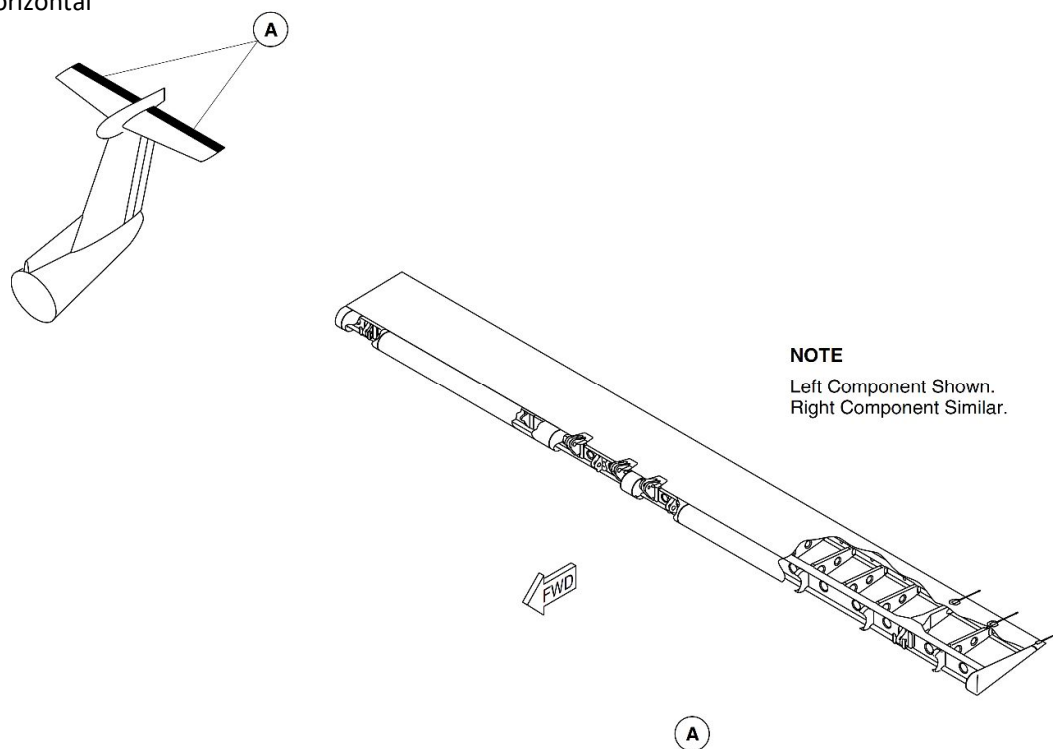


Figure 55-10 Elevators – Location

Detailed Description

[Refer Figure 55-11](#)

The left and right elevators each have a main and rear spar connected by profile ribs and covered by skin panels. The inboard half of the elevator also has a front spar fitted with a leading edge sealing strip.

The elevator are attached to the horizontal stabilizer, by four hinges using low torque plain roller bearings. There is an inboard and an outboard hinge fitting and two more hinge fittings near the centre of the front spar. The three PCU attachment fittings are located near the centre hinge points. The LVDT attachment fitting is located near the inboard PCU attachment point.

Box Structure

Each elevator has 28 ribs installed between the front and rear spars to make a box structure. The top and bottom skin panels are covers for the box structure. The skin panels are joined together aft of the rear spar by a spanwise trailing edge member. Each skin panel extends forward of the front spar but not fully over the elevator nose area. Static dischargers are attached to the trailing edge, near the outboard end of each elevator.

The spars, ribs, skins and trailing edge member are made from clad aluminum alloy sheet. The aluminum used to make the skins is chem-milled.

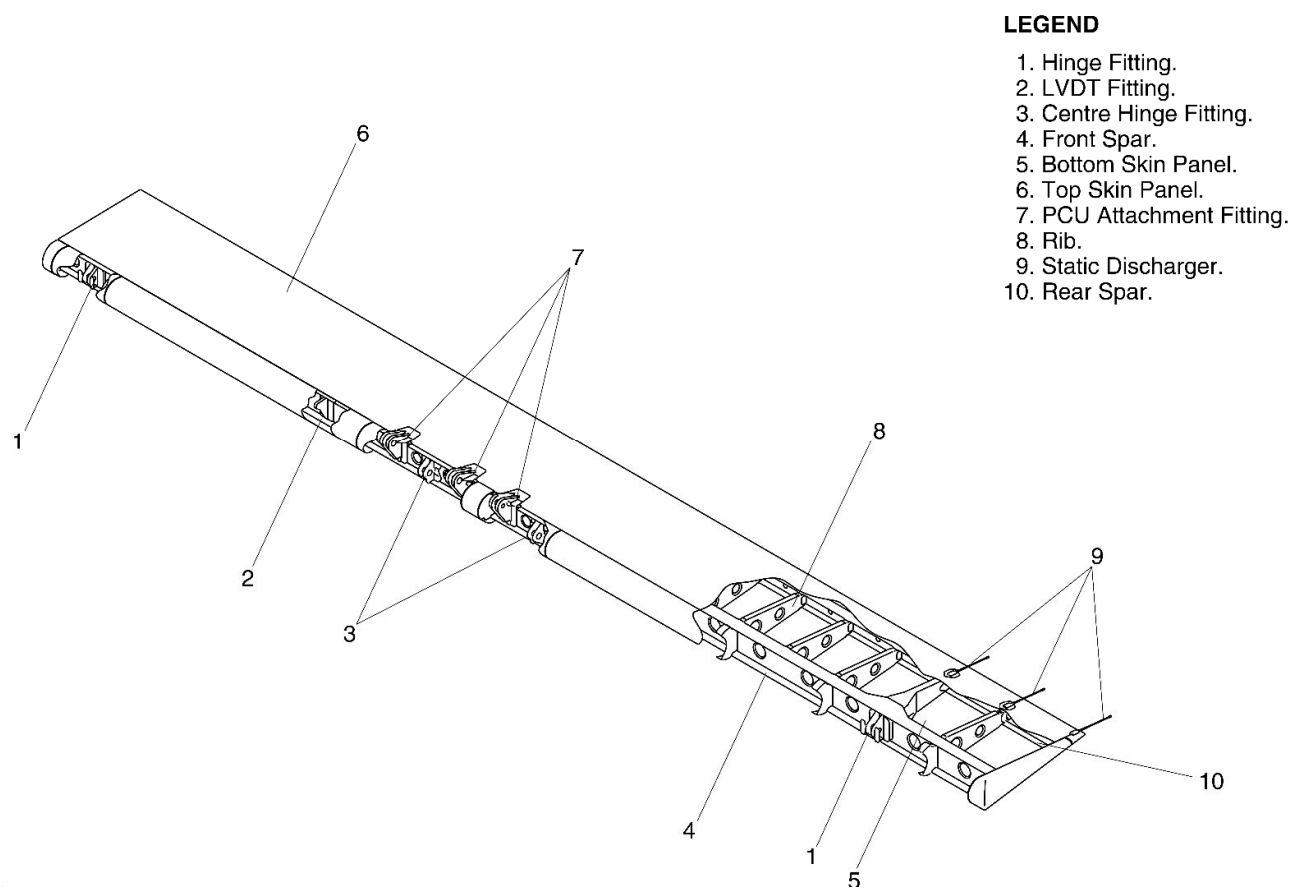


Figure 55-11 Elevators

Elevator Attachment

[Refer Figure 55-12](#) & [Refer Figure 55-13](#)

Each elevator is attached to the horizontal stabilizer by four hinge fittings. Each hinge fitting is made from aluminum alloy forgings. The hinge fittings are installed on the forward face of the front spar. The hinge fittings have forked lugs and are located: one at the inboard end, two in the centre and one at the outboard end.

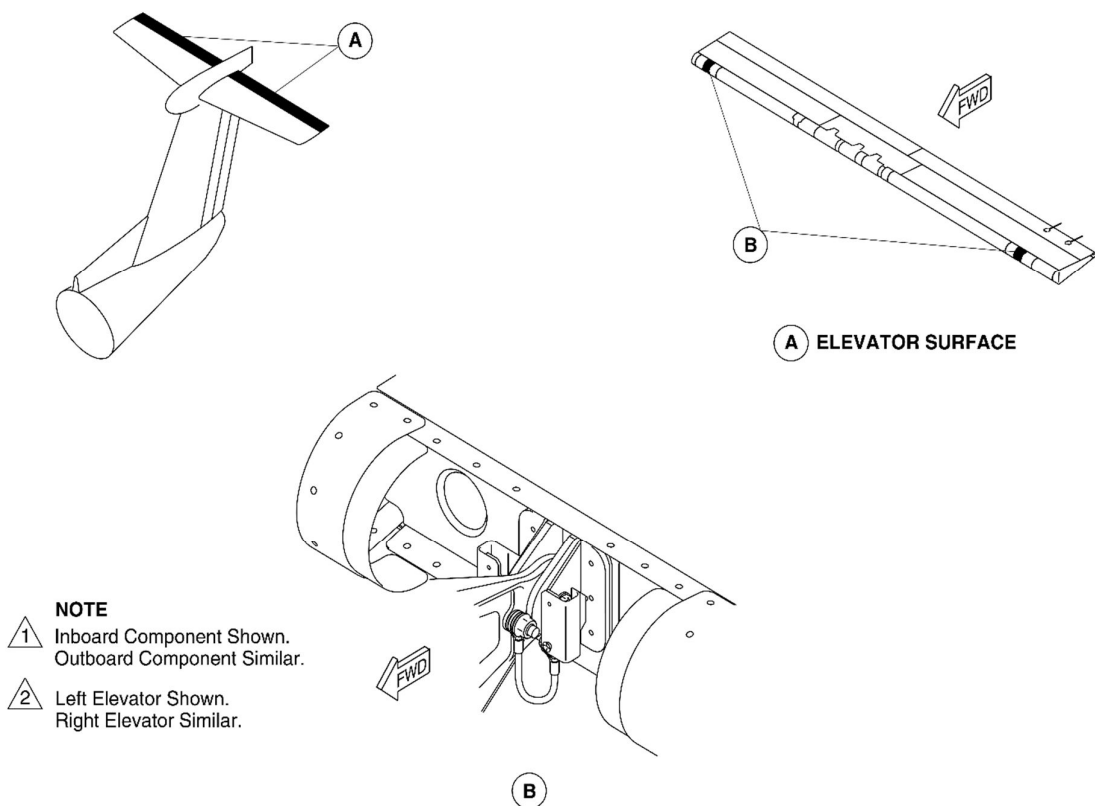


Figure 55-12 Elevators – Inboard / Outboard Elevator Hinges

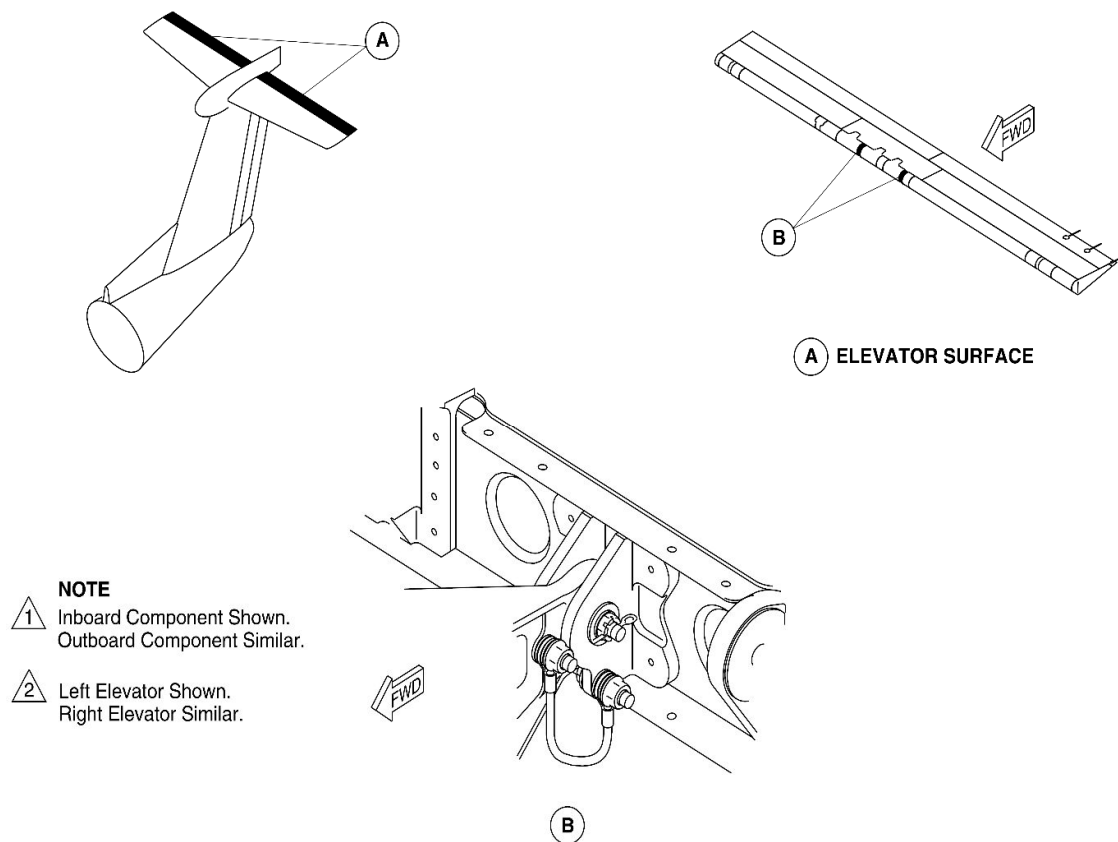


Figure 55-13 Elevators – Centre Hinges

Elevator PCU and LVDT Attachment

[Refer Figure 55-14](#)

Three PCU attachment fittings are located on the forward face of the front spar near the centre of the elevator. An LVDT attachment fitting is installed on the front spar just inboard of the PCU fittings. All the fittings are machined from aluminum alloy forgings.

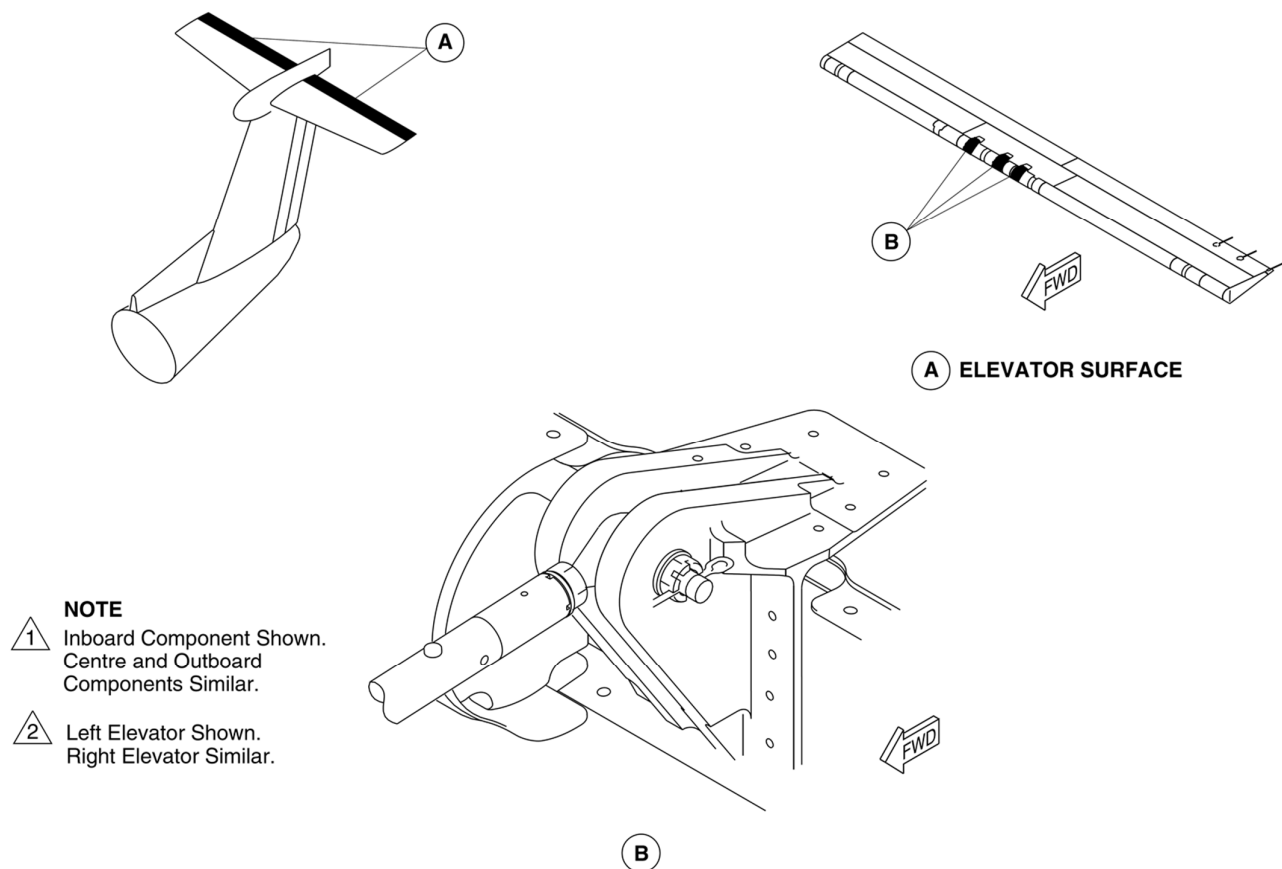


Figure 55-14 Elevators – PCU Fittings

Elevator Access Panels

[Refer Figure 55-15](#)

Five panels on the forward edge of each elevator give access to the hinge, actuator and LVDT attachment points. The access panels are made from clad aluminum alloy sheet.

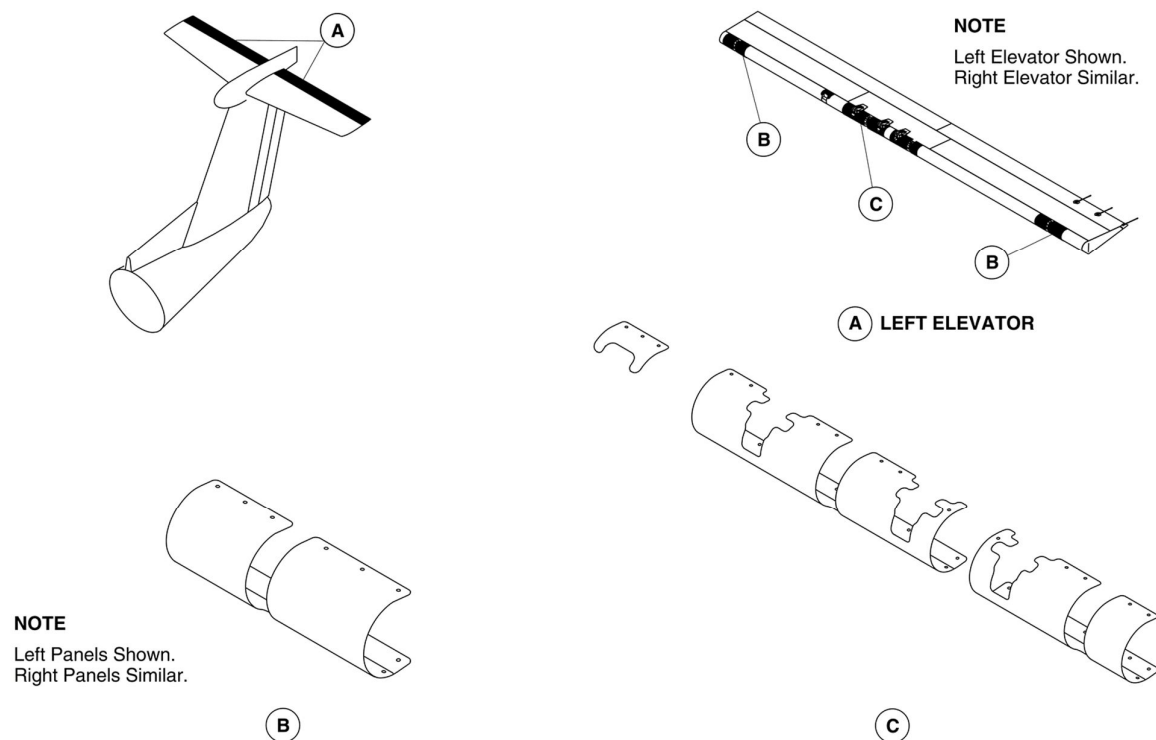


Figure 55-15 Elevators – Access Panels

Vertical Stabilizer

Introduction

The vertical stabilizer is an airfoil that stabilizes movement of the aircraft around the vertical (yaw) axis.

General Description

The vertical stabilizer has the components that follow:

- Vertical Stabilizer Access Panels
- Vertical Stabilizer Fairing
- Vertical Stabilizer Leading Edge

Detailed Description

The vertical stabilizer is part of the aft/tail fuselage. The stabilizer has three vertical spars connected by horizontal ribs and covered with skin panels. The vertical spar members are shaped at the lower end to form the main frames of the aft/tail fuselage. The leading edge has pneumatic deicer boots installed.

Vertical Stabilizer Access Panels

[Refer Figure 55-16](#)

There are five access panels on each side of the vertical stabilizer. The panels give access to the rudder and elevator control systems.

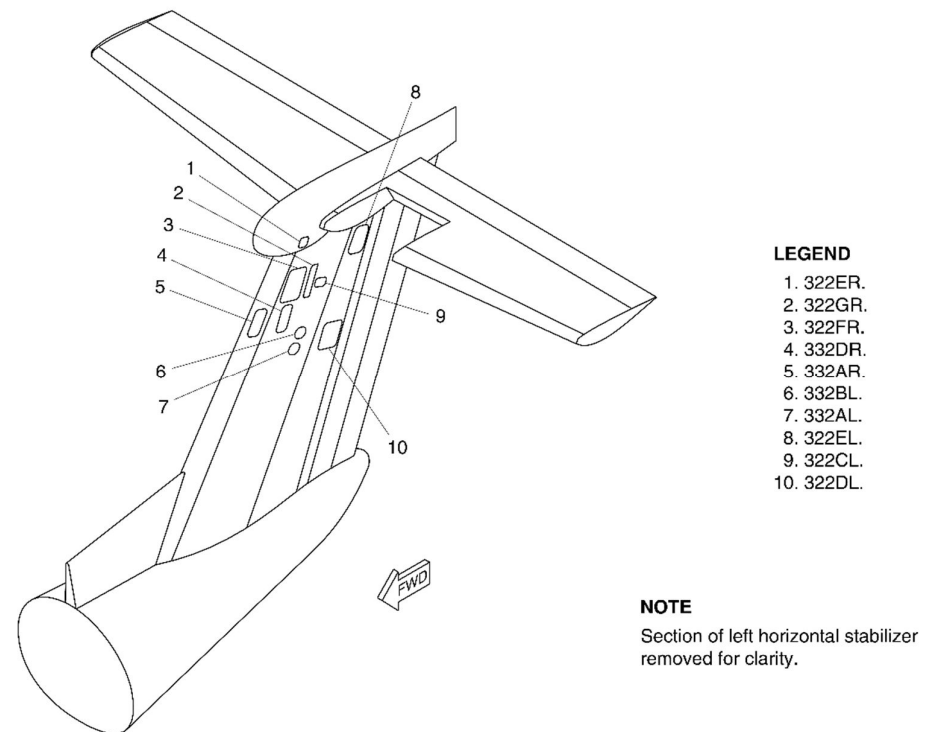


Figure 55-16 Vertical Stabilizers – Access Panels

Vertical Stabilizer Fairing

[Refer Figure 55-17](#)

The vertical stabilizer fairing is a combination of aerodynamic fairings. It starts at the top of the vertical stabilizer leading edge and continues aft to the rear of the horizontal stabilizer. The vertical stabilizer fairing is made from composite materials.

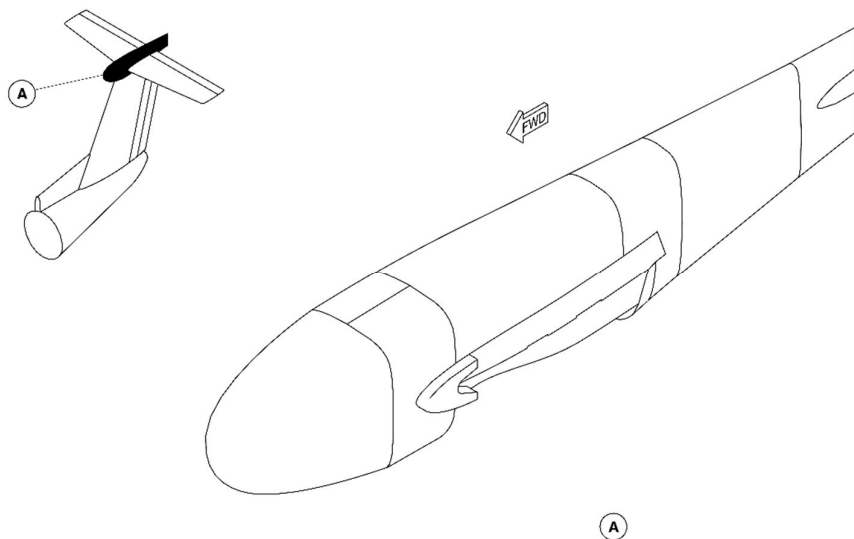


Figure 55-17 Vertical Stabilizer Fairing - Location

Vertical Stabilizer Leading Edge

[Refer Figure 55-18](#)

The vertical stabilizer leading edge is attached to the vertical stabilizer. It fairs with the aft dorsal fin at the lower end and with the vertical stabilizer fairing at the upper end.

The vertical stabilizer leading edge also contains the H.F. antenna. The vertical stabilizer leading edge is made from composite materials.

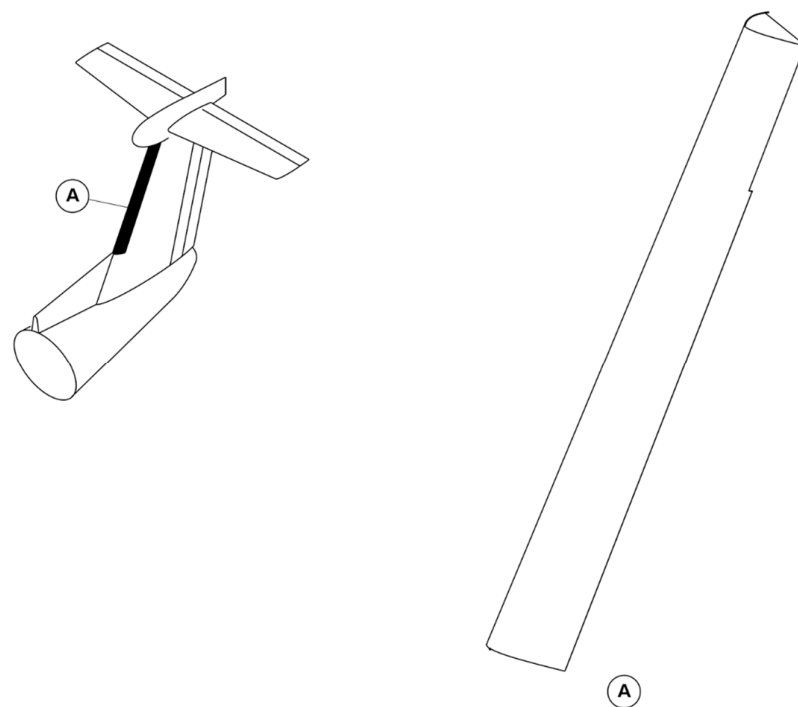


Figure 55-18 Vertical Stabilizer Leading Edge - Location

The Horizontal/Vertical Stabilizer Joint

[Refer Figure 55-19](#)

The horizontal/vertical stabilizer joint is installed in the center box of the horizontal stabilizer and front spar of the vertical stabilizer. The joint between the vertical and horizontal stabilizer is covered by a removable composite construction aerodynamic fairing.

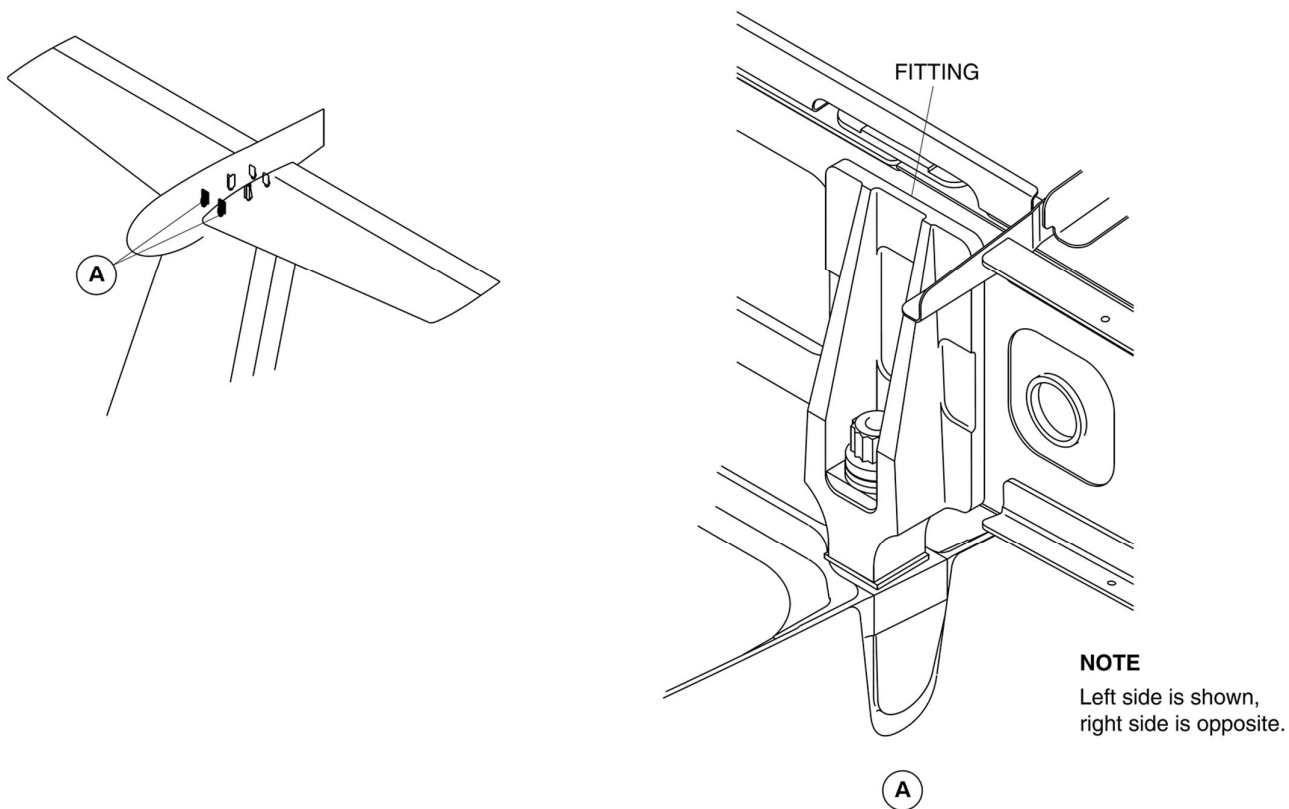


Figure 55-19 Horizontal/Vertical Stabilizer – Front Spar

Rudder

Introduction

The rudder is a moveable flight control surface that controls movement of the aircraft around the vertical axis.

General Description

The rudder is a primary flight control surface that is attached to the rear spar of the vertical stabilizer.

The rudder has the components that follow:

- Leading Edge
- Box Structure
- Rudder Attachment
- Rudder Access Panels

Detailed Description

[Refer Figure 55-20](#) & [Refer Figure 55-21](#)

The rudder has a fore rudder and a trailing rudder. The fore rudder is attached to the vertical stabilizer rear spar, and the trailing rudder is attached to the trailing edge of the fore rudder.

The fore rudder is moved directly by hydraulic power. Mechanical linkage transmits fore rudder movement to the trailing rudder. Both rudders are constructed of spars and profile ribs covered with skin panels.

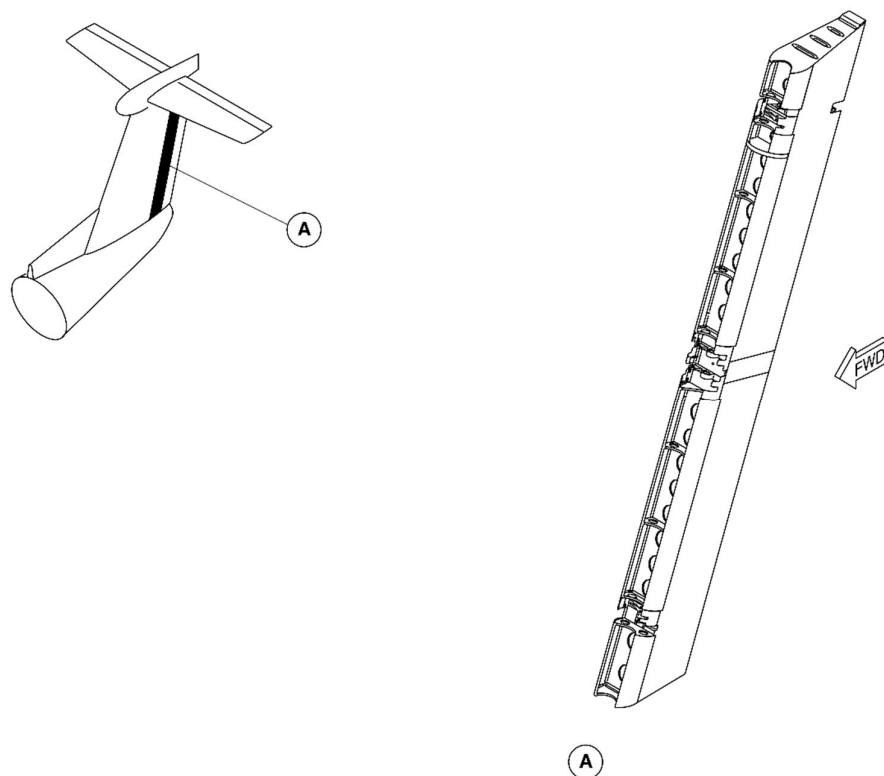


Figure 55-20 Fore Rudder– Location

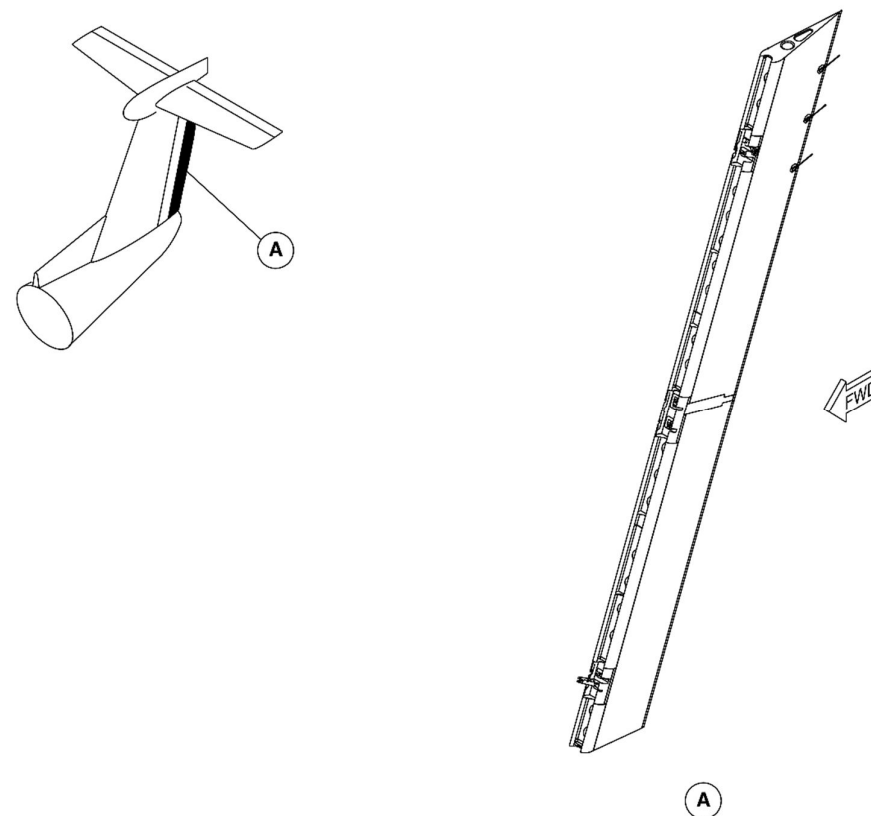


Figure 55-21 Trailing Rudder – Location

Leading Edge

[Refer Figure 55-22](#) & [Refer Figure 55-23](#)

The fore rudder leading edge section has 11 nose ribs attached to the forward face of the front spar. The trailing rudder leading edge section has 10 nose ribs attached to the forward face of the front spar. Upper and lower skin panels cover the leading edge section. The leading-edge skin panels do not fully close the nose area, they are cut away to give access to the attachment points.

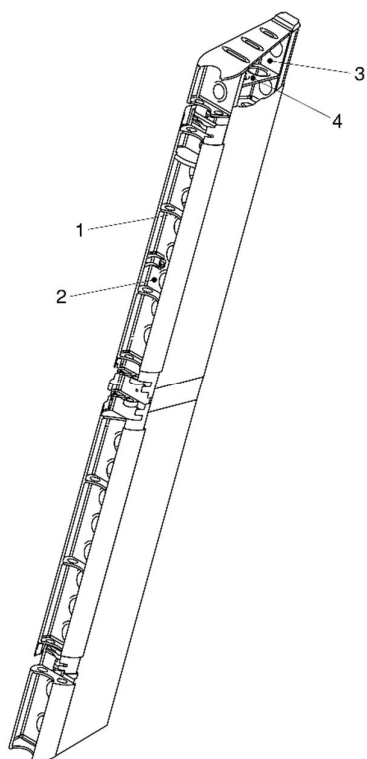
The leading edge parts are made from clad aluminum-alloy sheet.

Box Structure

The fore rudder box structure has 26 ribs attached to the front and rear spar assemblies. The trailing rudder box structure has 30 ribs attached to the front and rear spar assemblies. The top and bottom closing ribs are canted to align with the profile of the vertical stabilizer fairings and the tail cone. Left and right skin panels cover the box structure. Three static dischargers are installed near the top of the trailing rudder trailing edge.

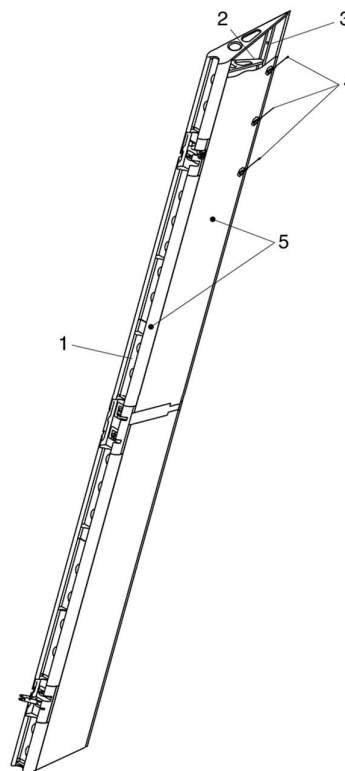
The front and rear spar assemblies are made from clad aluminum-alloy sheet and machined aluminum-alloy extrusions.

The ribs are made from clad aluminum-alloy sheet. The rib at the hinge locations are also reinforced with machined aluminum alloy members. The skin panels are made from clad aluminum-alloy sheet.



LEGEND

- 1. Nose rib.
- 2. Front spar.
- 3. Rear spar.
- 4. Rib.



LEGEND

- 1. Front spar.
- 2. Rib.
- 3. Rear spar.
- 4. Static discharger.
- 5. Skin panel.

Figure 55-22 Fore Rudder

Figure 55-23 Trailing Rudder

Rudder Attachment

[Refer Figure 55-24](#)

Six fittings on the fore rudder front spar form attachment points for the vertical stabilizer. Four of the fittings are hinges that are attached at the top and bottom of the spar. The fittings are machined from aluminum alloy forgings. The remaining fittings are combined hinge and actuator attachment

fittings, and are attached to the center of the spar. The fittings are machined from aluminum alloy forgings.

Six fittings on the fore rudder rear spar have attachment points for the trailing rudder. The fittings are hinges that are attached at the top, center and bottom of the spar. These fittings are machined from aluminum alloy forgings.

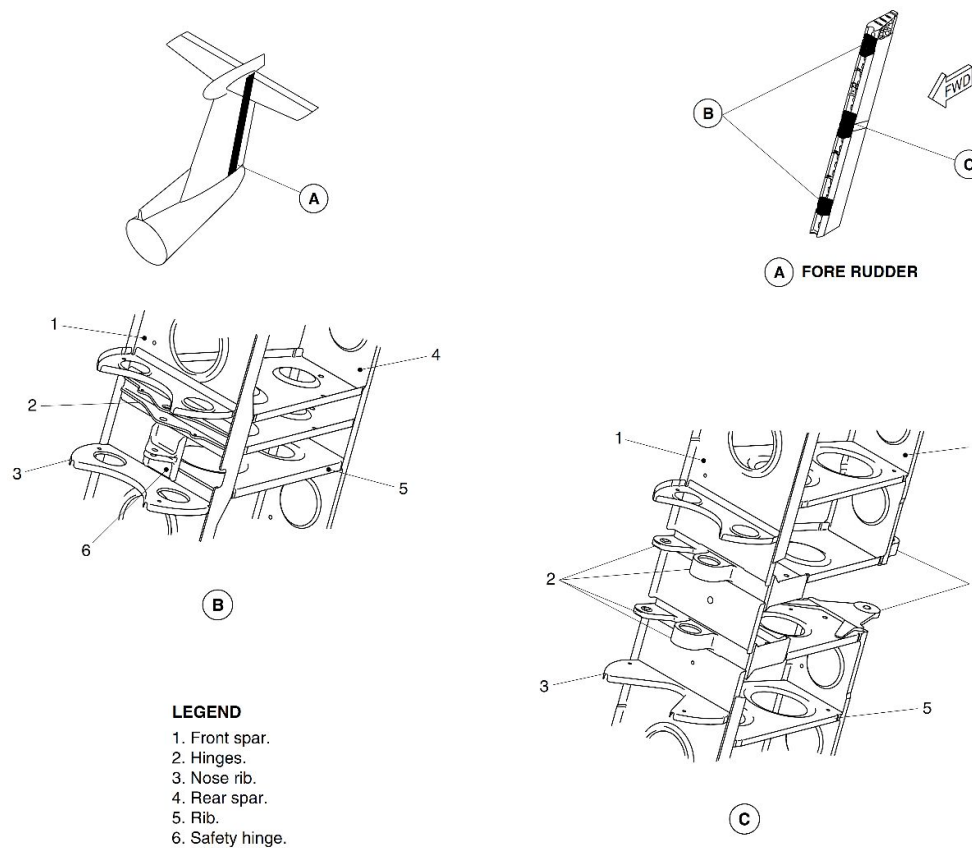


Figure 55-24 Fore Rudder Hinges

Refer Figure 55-25

Six fittings on the trailing rudder front spar have attachment points for the fore rudder. Four of the fittings are hinges that are attached at the top, center and bottom of the spar. The fittings are machined from aluminum alloy

forgings. The remaining fittings are combined hinge and radius rod attachment fittings, and are attached at the top and bottom of the spar. The fittings are machined from aluminum alloy forgings.

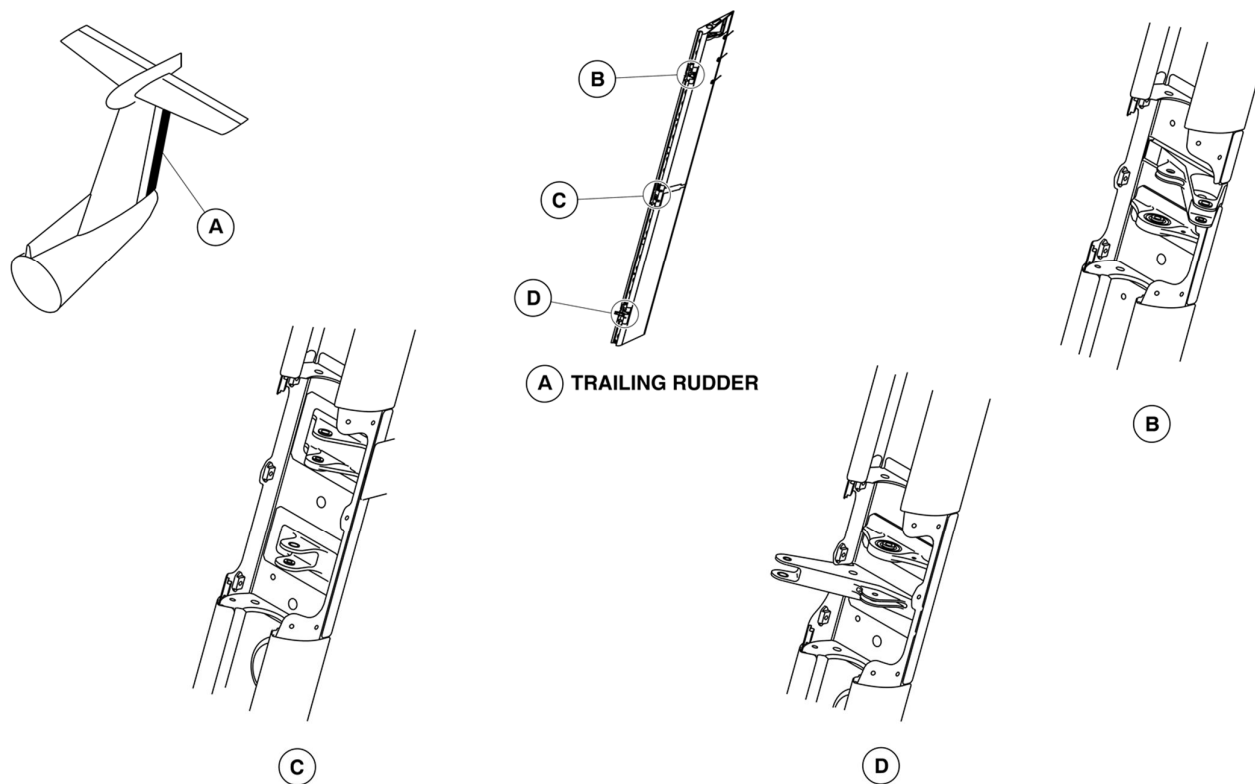


Figure 55-25 Trailing Rudder Hinges

Rudder Access Panels

[Refer Figure 55-26](#) & [Refer Figure 55-27](#)

There are four access panels on the left and right side of each rudder leading edge. The panels give access to the rudder hinge and PCU attachments.

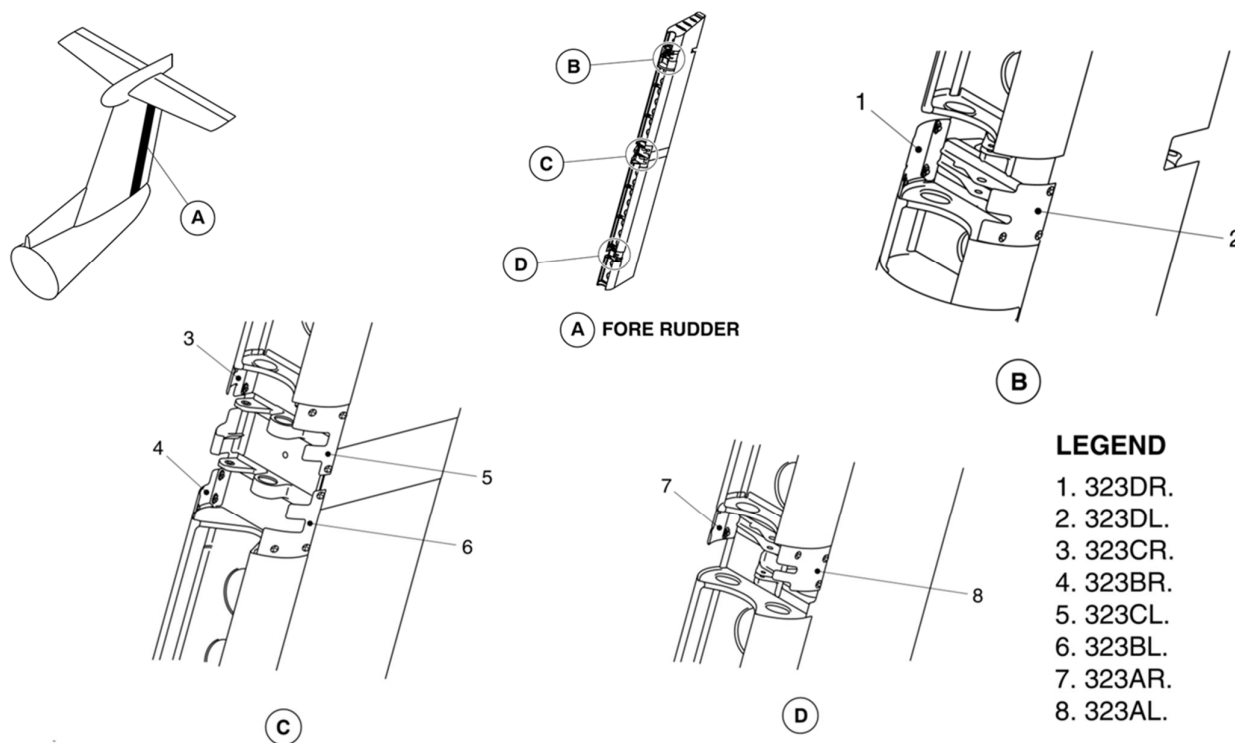


Figure 55-26 Fore Rudder Access Panels

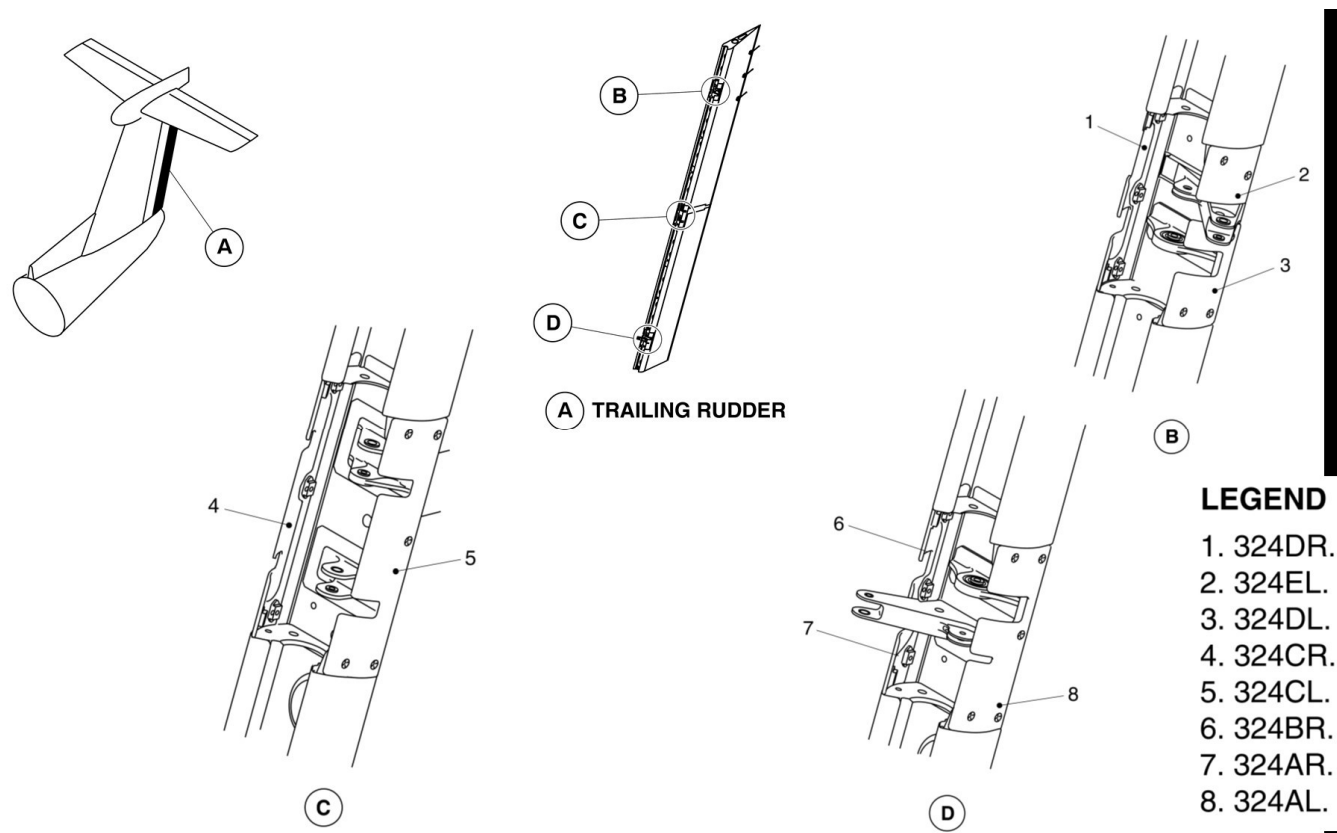


Figure 55-27 Trailing Rudder Access Panels